

Bipin Kumar Chaudhary

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Portfolio | LinkedIn | GitHub | LeetCode

SUMMARY

Full Stack Engineer with **3+ years** of experience designing and shipping production systems end-to-end from serverless video pipelines on AWS to AI-enriched workflows and real-time scheduling engines. M.Tech (CSE) from MNNIT Allahabad with **350+ LeetCode** problems solved.

EXPERIENCE

- Self-Employed** | Independent Full-Stack Developer | *Remote* Oct 2023 – Present
- Built production-grade video streaming and clinic management platforms with end-to-end architecture: auth, scheduling engines, caching, and async workflows.
 - Integrated WhatsApp API and OpenAI for real-time communication and AI-powered features; active open-source contributor.
- SAP Labs** | Graduate Research Engineer | *iXp Internship, M.Tech MNNIT Allahabad | Bangalore* Aug 2024 – Jun 2025
- Identified **60+ bugs** across SAP Business Application Studio via functional and regression testing, with detailed defect reports to accelerate triage.
 - Developed **30+ automated test scripts** and **50+ BAS applications**, converting manual test cases into repeatable validation pipelines.
 - Validated HANA Vector Engine and database artifacts across **100+ test scenarios** with zero critical failures at production sign-off.
- Nagarro** | Full Stack Developer | *Gurgaon* May 2021 – Sep 2023
- Reduced frontend TBT by **20%** using Angular OnPush strategy and reusable components, sustaining **60 FPS** under high data load.
 - Built scalable backend services for a rail asset management system handling **170K+ assets** with sub-second real-time updates.
 - Improved deployment speed by **15%** by standardizing CI/CD pipelines and Jasmine unit testing; authored team-reference API documentation.

PROJECTS

- StreamSphere** | *Angular, Node.js, AWS, MongoDB, Redis* [Live] | [GitHub]
- Built a **serverless video pipeline** (S3 → Lambda → FFmpeg → HLS) with adaptive bitrate streams (360p/720p/1080p), cutting playback startup time by **40%**.
 - Implemented **event-driven Redis caching** with Lambda webhooks, reducing feed latency by **30–40%** and S3 origin load by **25%**.
 - Designed **cursor-based pagination** for infinite scroll with O(1) query performance and consistency under concurrent uploads.
 - Built a fault-tolerant **AI enrichment pipeline** (Whisper, GPT, HuggingFace) with async orchestration and graceful degradation.
- DentalConnect** | *Next.js, PostgreSQL (Supabase), Vercel, WhatsApp API* [Live] | [GitHub]
- Designed full-stack appointment platform with OTP-based WhatsApp auth, JWT role-based access, and a **dynamic scheduling engine** reconciling live bookings in real time.
 - Built **async notification workflows** (OTP, reminders, cancellations) via WhatsApp Business API and Vercel Cron, with RLS, Zod validation, and OTP rate limiting.
- Argument** | *React, Node.js, Socket.io, MongoDB, Redis, OpenAI, Whisper* [Live] | [GitHub]
- Engineered a **real-time debate platform** with Socket.io supporting room codes, alternate-turn and buzzer modes, and a host-controlled session lifecycle.
 - Built a **live transcription pipeline** using Whisper API feeding transcripts into an OpenAI **scoring engine** that evaluates arguments and determines round outcomes.
 - Designed a **gamification system** with XP progression, dynamic leaderboards, and badge unlocks; Redis-backed session tokens for low-latency room auth.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, C++, Java, SQL
Core Concepts: Data Structures & Algorithms, System Design, OOP, Concurrency, Caching, Distributed System, REST APIs
Frontend: Angular, React, Next.js, HTML5/CSS3, Tailwind CSS
Backend: Node.js, Express, REST APIs, CI/CD, Docker, Git
AI/ML: OpenAI APIs, Whisper (ASR), HuggingFace Transformers, RAG, Vector Embeddings
Cloud & Databases: AWS (S3, Lambda, CloudFront), SAP BTP, SAP HANA, MongoDB, Redis, PostgreSQL

EDUCATION

- Motilal Nehru National Institute of Technology Allahabad** | M.Tech(CSE) | **CPI: 8.75/10**
Includes SAP Labs iXp internship (Aug 2024 – Jun 2025) Aug 2023 – Aug 2025
- Bundelkhand Institute of Engineering and Technology** | B.Tech(CSE) | **78.6%** Aug 2017 – Aug 2021